



Question Answering System in Vietnamese Legal Law

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01. INTRODUCTION

In today's world of rapidly increasing data volumes, the ability to **accurately retrieve** and **recommend** relevant documents has become essential for enhancing the reliability and effectiveness of Question Answering (QA) systems. This is especially true in the **legal domain**, where precise and contextually appropriate information retrieval is critical. Recent advancements, such as Retrieval Augmented Generation (**RAG**), have garnered significant attention for their ability to enhance the capabilities of large language models (LLMs) by **reducing hallucination** issues within QA systems.

PROBLEM STATEMENT

- Existing retrieval methods (e.g., dense embeddings, hybrid techniques) face **limitations in Vietnamese legal contexts**. Common issues include:
 - Inefficient data processing causing long token lengths.
 - Slow retrieval times affecting system performance.
 - Naive combination of retrieval results reducing accuracy.
- A key overlooked issue is the **ordering of retrieved documents** before passing to LLMs.
=> Improper ordering can reduce answer quality and reliability in legal QA systems

OBJECTIVES

- •
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 - •
- Enhance Vietnamese legal information retrieval without fine-tuning or developing new models.
 - Use hybrid methods (keyword + semantic).
 - Apply reranking to prioritize the most relevant legal texts.
 - Ensure all techniques are optimized for Vietnamese law and language.

FEATURES

- Advanced RAG Architecture - Combines vector search, BM25, and cross-encoder reranking for optimal document retrieval
- Hybrid Search - Uses both semantic similarity (vector search) and keyword matching (BM25) to find relevant documents
- Question Refinement - Improves query understanding through automatic question refinement
- Cross-Encoder Reranking - Employs cross-encoder reranking to improve the accuracy of retrieved documents.
- Fallback Mechanisms - Integrates with Google Search to provide answers when legal documents are insufficient.
- Vietnamese-Optimized - Specifically designed for Vietnamese language processing and legal terminology

02. PROPOSED METHOD



WORKFLOW

Workflow of this project.



DATA COLLECTION

Data collection and preprocessing.



RAG IN LEGAL DOMAIN

What is RAG and RAG in this system.



MAIN FEATURES

Six main features.



2.1. WORKFLOW

The Vietnamese Legal Chatbot follows a modular workflow consisting of several key stages:

Legal documents are **collected** from official sources and processed through a **text cleaning** and **chunking pipeline**

→ These chunks are then **transformed** into **embeddings** or **keyword indexes** and stored in a **vector database (Qdrant)** along with metadata

→ When a user submits a natural-language query, the system retrieves the most relevant legal text segments using either **BM25** or **semantic search**. The retrieved information, combined with the user's question, is passed to a large language model (e.g., Gemini) in a retrieval-augmented generation (**RAG**) process

→ The model generates a response based strictly on the provided legal content, ensuring grounded and accurate answers

→ The final response, along with cited legal sources, is returned via an API or displayed through a user-friendly interface.

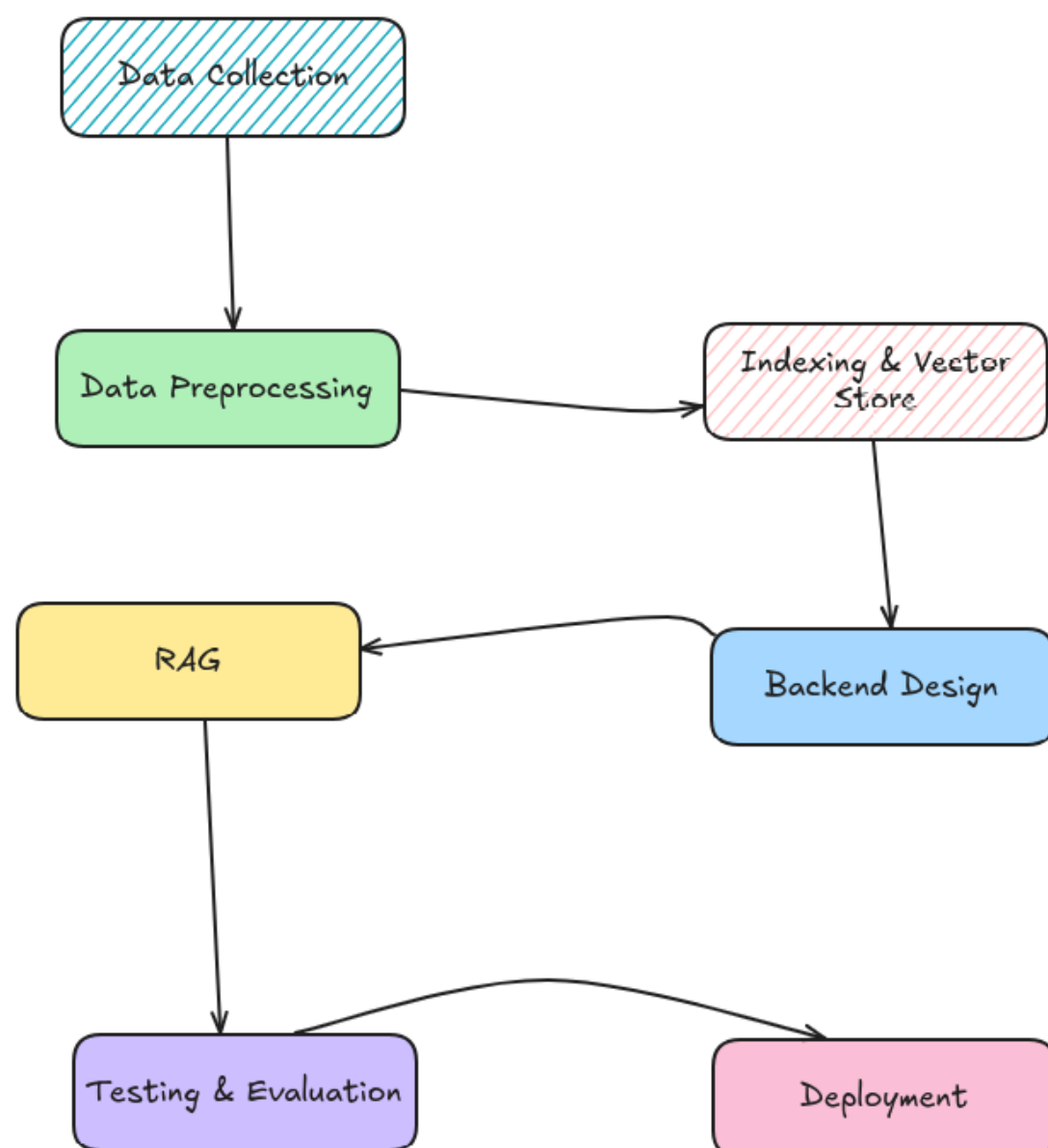


Figure 2.1. Workflow Diagram

2.2. RAG IN LEGAL DOMAIN

WHAT IS RAG ?



RAG is a **method** that combines three components:

- **Retriever**: Searches for relevant information from a knowledge base (e.g., documents, articles).
- **Augmentor**: Adds the found information to the input for the language model.
- **Generator**: Uses a language model (e.g., GPT, Gemini) to generate responses based on retrieved content.

It allows the model to "**look up**" information instead of relying only on what it was trained on.

WHY RAG ?



Reduces Hallucinations: Grounds answers in real documents, improving factual accuracy.

Stays Updated: Retrieves the latest info, unlike static models.

Improves Context: Adds relevant content to enhance conversation flow.

Handles Specialized Domains: Brings in expert knowledge from external sources.

Saves Cost: No need to fine-tune large models — just update the data.

Scales Easily: Works across domains like law, healthcare, finance.

2.2. RAG IN LEGAL DOMAIN

RAG IN MY PROJECT



Naive RAG may not be accurate enough. In this study, I used and compared the effectiveness of 4 types of RAG:

- Naive RAG (Vector-based RAG - Semantic RAG)
- Lexical RAG (Keyword-based RAG)
- Hybrid RAG (Combine both)
- Hybrid + Reranking



2.3. DATA COLLECTION

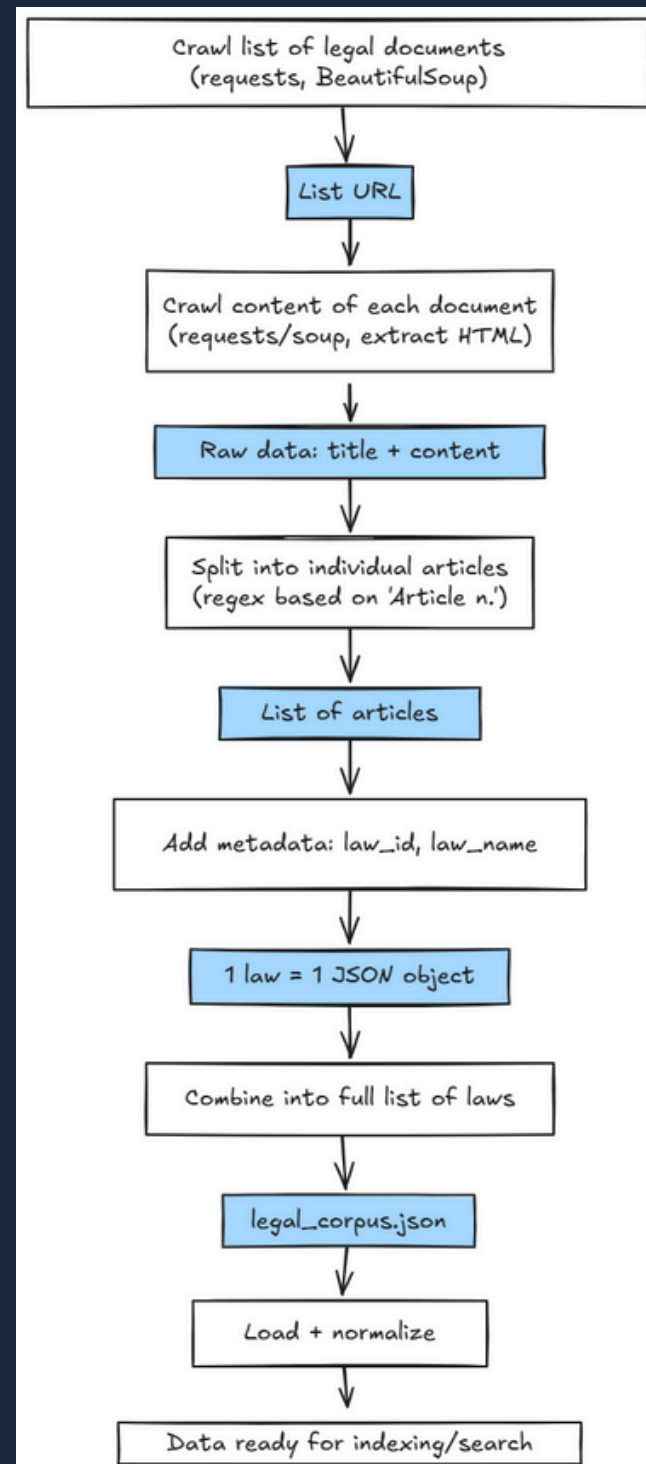


Figure 2.3.1. Cong Thong Tin Dien Tu Phap Dien - Data Source 1



Figure 2.3.2. Co So Du Lieu Quoc Gia - Data Source 2

2.3. DATA COLLECTION



```

"law_id": "01/2009/tt-bnn",
"articles": [
  {
    "article_id": "1",
    "title": "Điều 1. Phạm vi áp dụng",
    "text": "Thông tư này hướng dẫn tuần tra, canh gác bảo vệ đề Điều t
  },
  {
    "article_id": "2",
    "title": "Điều 2. Tổ chức lực lượng",
    "text": "1. Hàng năm trước mùa mưa, lũ, Ủy ban nhân dân cấp xã nơi
  },
  {
    "article_id": "3",
    "title": "Điều 3. Tiêu chuẩn của các thành viên thuộc lực lượng tuần
    "text": "1. Là người khỏe mạnh, tháo vát, đủ khả năng đảm đương nhữ
  },
  {
    "article_id": "4",
    "title": "Điều 4. Nhiệm vụ của lực lượng tuần tra, canh gác đề",
    "text": "1. Chấp hành sự phân công của Ban chỉ huy phòng, chống lự
  },
  {
    "article_id": "5",
    "title": "Điều 5. Phù hiệu của lực lượng tuần tra, canh gác đề",
    "text": "Phù hiệu của lực lượng tuần tra, canh gác đề là một băng đ
  },
]
  
```

convert →

```

law title,article_id,article_title,clause_id,point_id,sentence_id,content,conte
01/2009/tt-bnn,1,Điều 1. Phạm vi áp dụng,0,0,0,"thông tư này hướng dẫn tuần tra
01/2009/tt-bnn,2,Điều 2. Tổ chức lực lượng,1,1,0," khoản 1. hàng năm trước mùa
01/2009/tt-bnn,2,Điều 2. Tổ chức lực lượng,2,1,0," khoản 2. lực lượng tuần tra,
01/2009/tt-bnn,2,Điều 2. Tổ chức lực lượng,3,1,0," khoản 3. khi lũ, bão có diễn
01/2009/tt-bnn,3,Điều 3. Tiêu chuẩn của các thành viên thuộc lực lượng tuần tr
01/2009/tt-bnn,3,Điều 3. Tiêu chuẩn của các thành viên thuộc lực lượng tuần tr
01/2009/tt-bnn,4,Điều 4. Nhiệm vụ của lực lượng tuần tra, canh gác đề",1,1,0,"
01/2009/tt-bnn,4,Điều 4. Nhiệm vụ của lực lượng tuần tra, canh gác đề",2,1,0,"
01/2009/tt-bnn,4,Điều 4. Nhiệm vụ của lực lượng tuần tra, canh gác đề",3,1,0,"
01/2009/tt-bnn,4,Điều 4. Nhiệm vụ của lực lượng tuần tra, canh gác đề",4,1,0,"
01/2009/tt-bnn,4,Điều 4. Nhiệm vụ của lực lượng tuần tra, canh gác đề",5,1,0,"
01/2009/tt-bnn,5,Điều 5. Phù hiệu của lực lượng tuần tra, canh gác đề",0,0,0,"
01/2009/tt-bnn,6,Điều 6. Trang bị dụng cụ, số sách",1,1,0," khoản 1. lực lượng
01/2009/tt-bnn,6,Điều 6. Trang bị dụng cụ, số sách",2,1,0," khoản 2. số lượng
  
```

Figure 2.3.4. Sample Data (csv)

Figure 2.3.4. Sample Data (json)

I convert legal_corpus.json to legal_sentences.csv for easier viewing, debugging, or manual inspection. The implementation (indexing, retrieval, and chatbot) still uses the original structured JSON to preserve legal hierarchy and relationships between laws, articles, clauses, and sentences.

Figure 2.3.3. Crawl Data Flow

2.4. MAIN FEATURES

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-
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ADVANCED RAG ARCHITECTURE

Combines vector search, BM25, and cross-encoder reranking for optimal document retrieval



QUESTION REFINEMENT

Improves query understanding through automatic question refinement



HYBRID SEARCH

Uses both semantic similarity (vector search) and keyword matching (BM25) to find relevant documents



CROSS-ENCODER RERANKING

Employs cross-encoder reranking to improve the accuracy of retrieved documents



FALLBACK MECHANISMS

Integrates with Google Search to provide answers when legal documents are insufficient



VIETNAMESE-OPTIMIZED

Specifically designed for Vietnamese language processing and legal terminology

2.4. MAIN FEATURES



ADVANCED RAG ARCHITECTURE

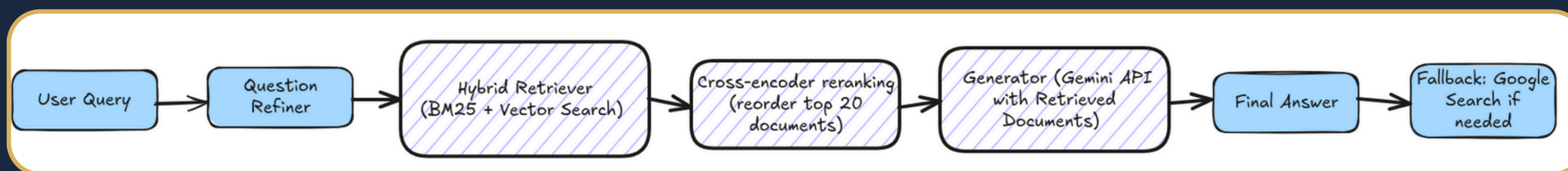


Figure 2.4.1. Advanced RAG Architecture

-  **Hybrid Search:** Combines semantic (vector) and keyword (BM25) matching.
-  **Cross-Encoder Reranking:** Improves relevance of retrieved docs.
-  **Question Refinement:** Auto-improves vague queries.
-  **Fallback Mechanism:** Uses Google if no document found.
-  **Vietnamese-Optimized:** Tailored chunking, tokenization, legal terms.

SYSTEM ARCHITECTURE

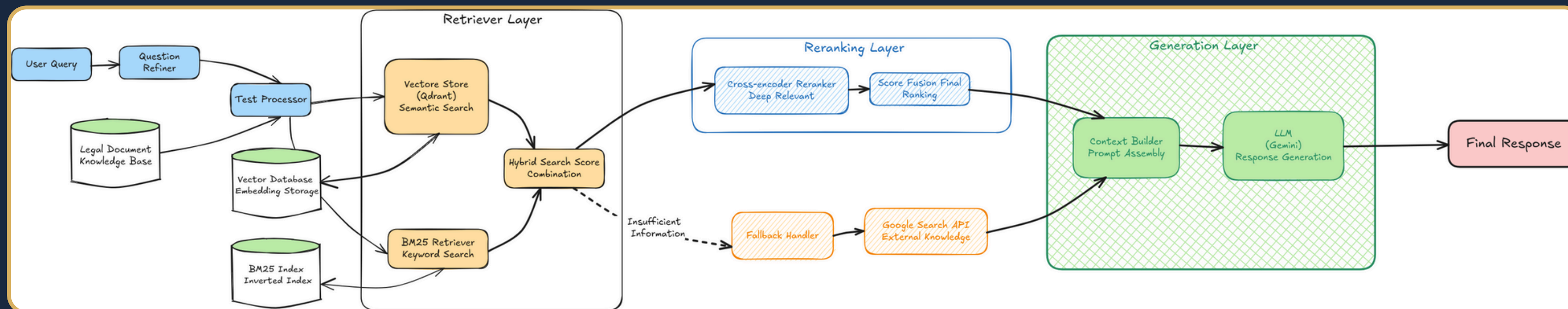


Figure 2.4.2. Advanced RAG Architecture

The system follows a modern RAG architecture with three primary layers:

Retrieval Layer

- Vector Store (Qdrant) - Semantic search using dense vector embeddings
- BM25 Retriever - Statistical keyword-based search
- Hybrid Search - Combines and deduplicates results from both retrieval methods

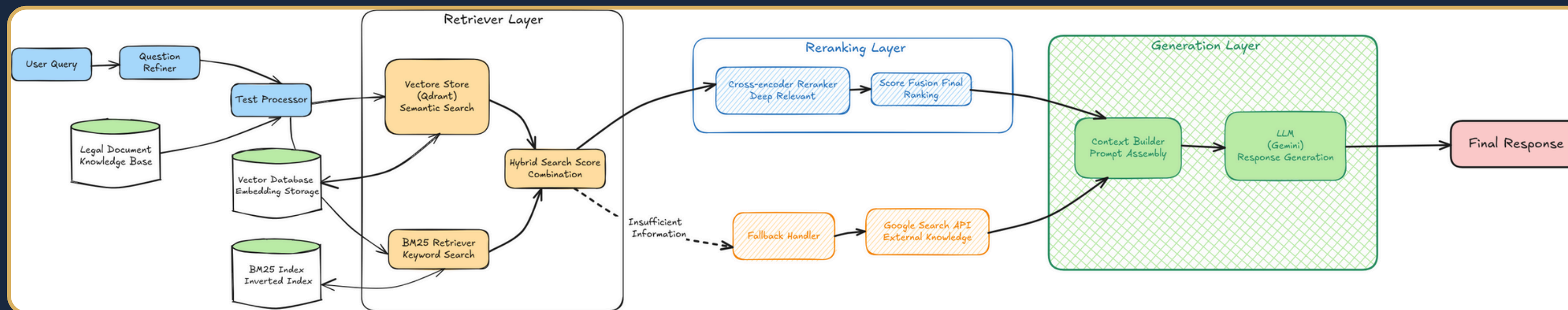
Reranking Layer

- Cross-Encoder Reranker - Precisely scores document-query pairs for relevance
- Score Fusion - Intelligently combines original retrieval scores with reranker scores

Generation Layer

- Context Builder - Formats retrieved documents into a prompt context
- LLM (Gemini) - Generates natural language responses based on the retrieved context

SYSTEM ARCHITECTURE



1. Legal Document Knowledge Base

- **Role:** Stores the original legal documents in raw or structured form.
- **Input to:** Text Processor and BM25 Index, which process these documents into searchable formats.
- **Data Type:** Full-text legal content (e.g., laws, decrees, circulars).

2. BM25 Index (Inverted Index)

- **Role:** Stores documents in a keyword-indexed format for traditional lexical search.
- **Used by:** BM25 Retriever, which retrieves relevant documents based on keyword overlap with the query.
- **Technology:** Qdrant BM25

3. Vector Database / Embedding Storage

- **Role:** Stores dense embeddings (vectors) of document chunks, used for semantic search.
- **Used by:** Vector Store (Qdrant) to retrieve top-k semantically similar chunks.
- **Key Characteristics:**
 - Uses dense vector embeddings
 - Converts both the query and documents into vectors (lists of numbers that represent meaning)
 - Then compares them using similarity (like cosine similarity)

2.3. MAIN FEATURES



QUESTION REFINEMENT

- Improves document retrieval by aligning vague queries with actual legal terminology.
- Enhances both recall and precision in hybrid search.

Step-by-Step Refinement:

1. **Basic Cleaning:** "quyen va nghĩa vụ của nld là gì?"
2. **Spelling Correction:** "quyền và nghĩa vụ của nld là gì?"
3. **Abbreviation Expansion:** "quyền và nghĩa vụ của nld người lao động là gì?"
4. **Context Detection:** ["labor"]
5. **Intent Analysis:** "quyen va nghĩa vụ của nld là gì?" (json) (goal/ purpose)
6. **Context Enhancement:** "quyền và nghĩa vụ của nld người lao động là gì? luật lao động bảo hiểm xã hội"
7. **Advanced Refinement:** "Quyền và nghĩa vụ của người lao động theo quy định của pháp luật lao động Việt Nam là gì?"
8. **Validation:**  PASSED

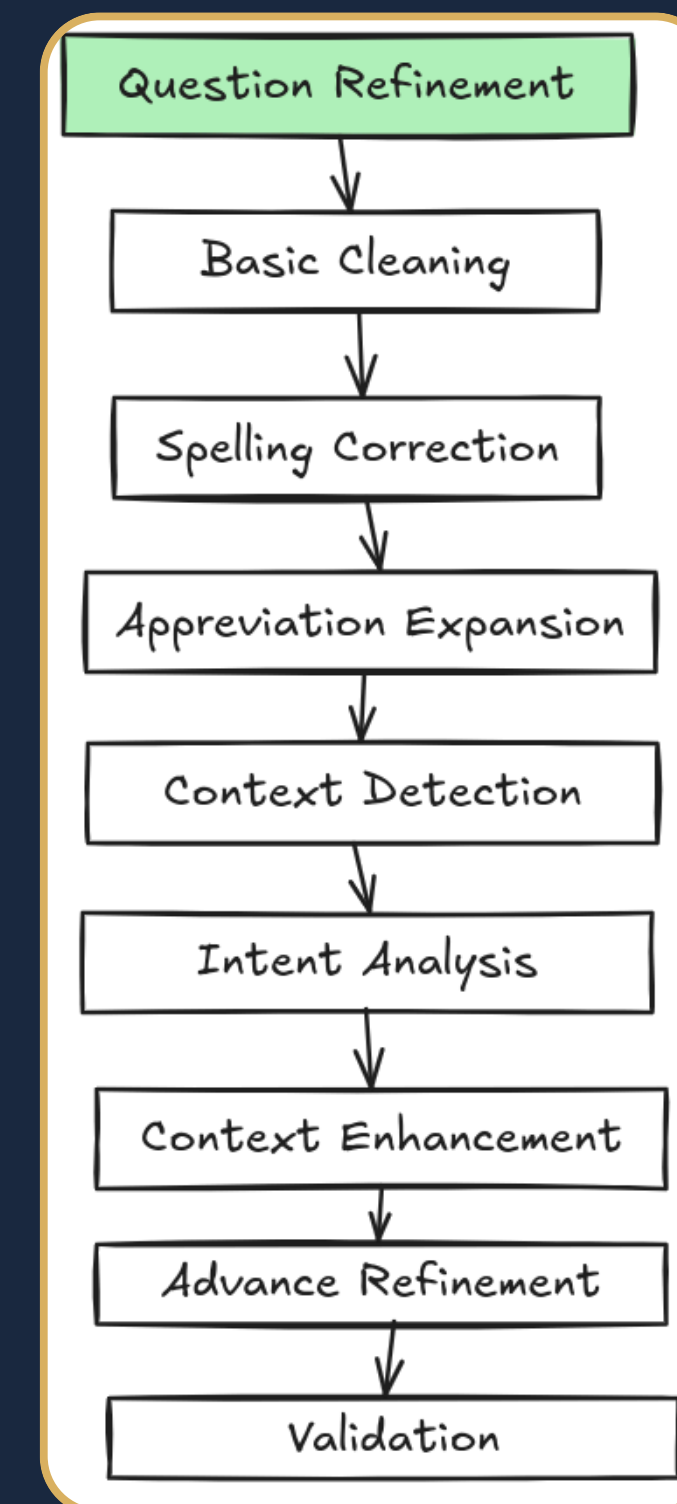


Figure 2.4.3. Question Refinement

2.4. MAIN FEATURES



HYBRID SEARCH

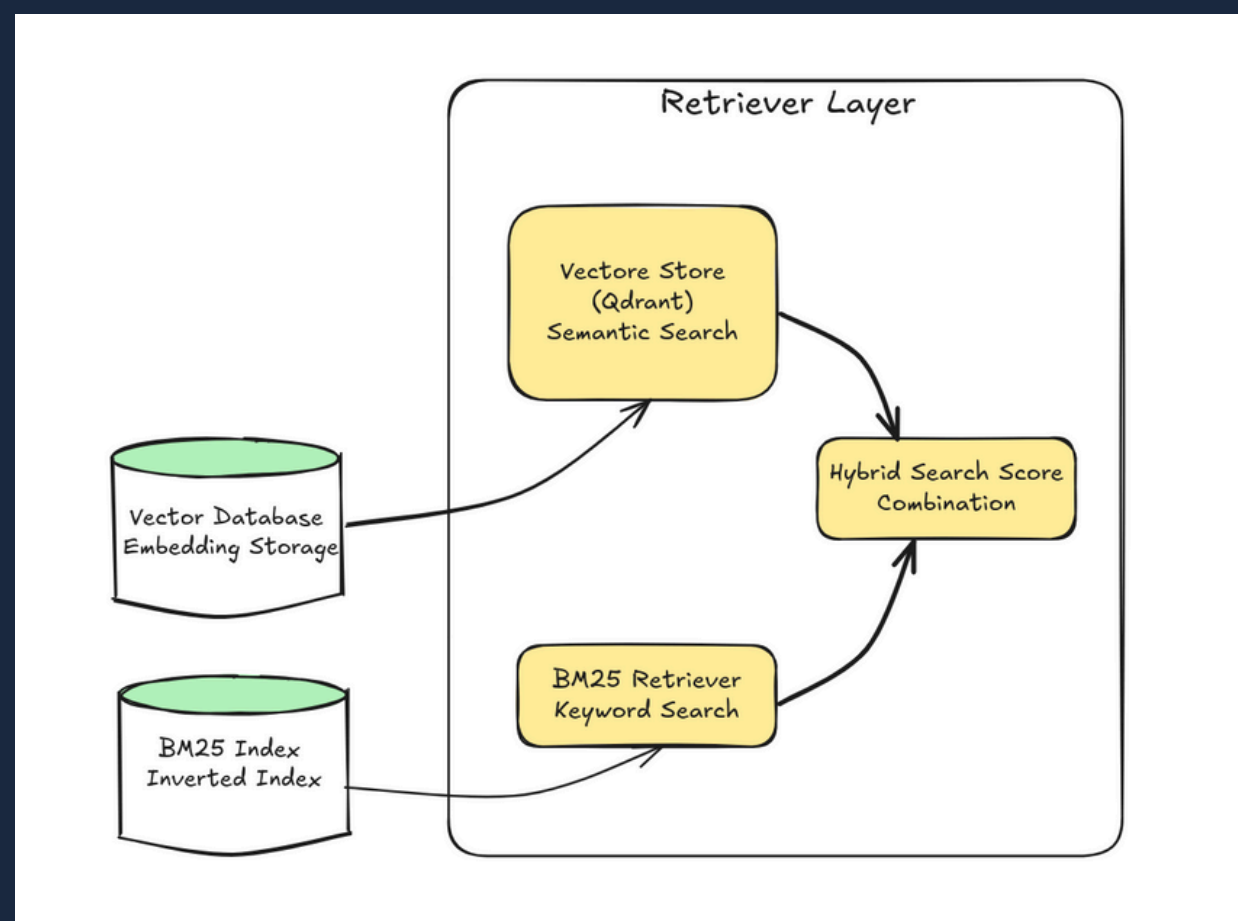


Figure 2.4.4. Hybrid Search

Vector Search: Finds documents similar in meaning, even if different words are used.

BM25: Finds documents with exact keyword matches, useful for legal terms.

Combining Both: Increases both recall (semantic matches) and precision (exact matches).

→ *Helps capture queries like:*

"quyền nuôi con" → finds both "chăm sóc trẻ" and "quyền giám hộ"

2.4. MAIN FEATURES



HYBRID SEARCH

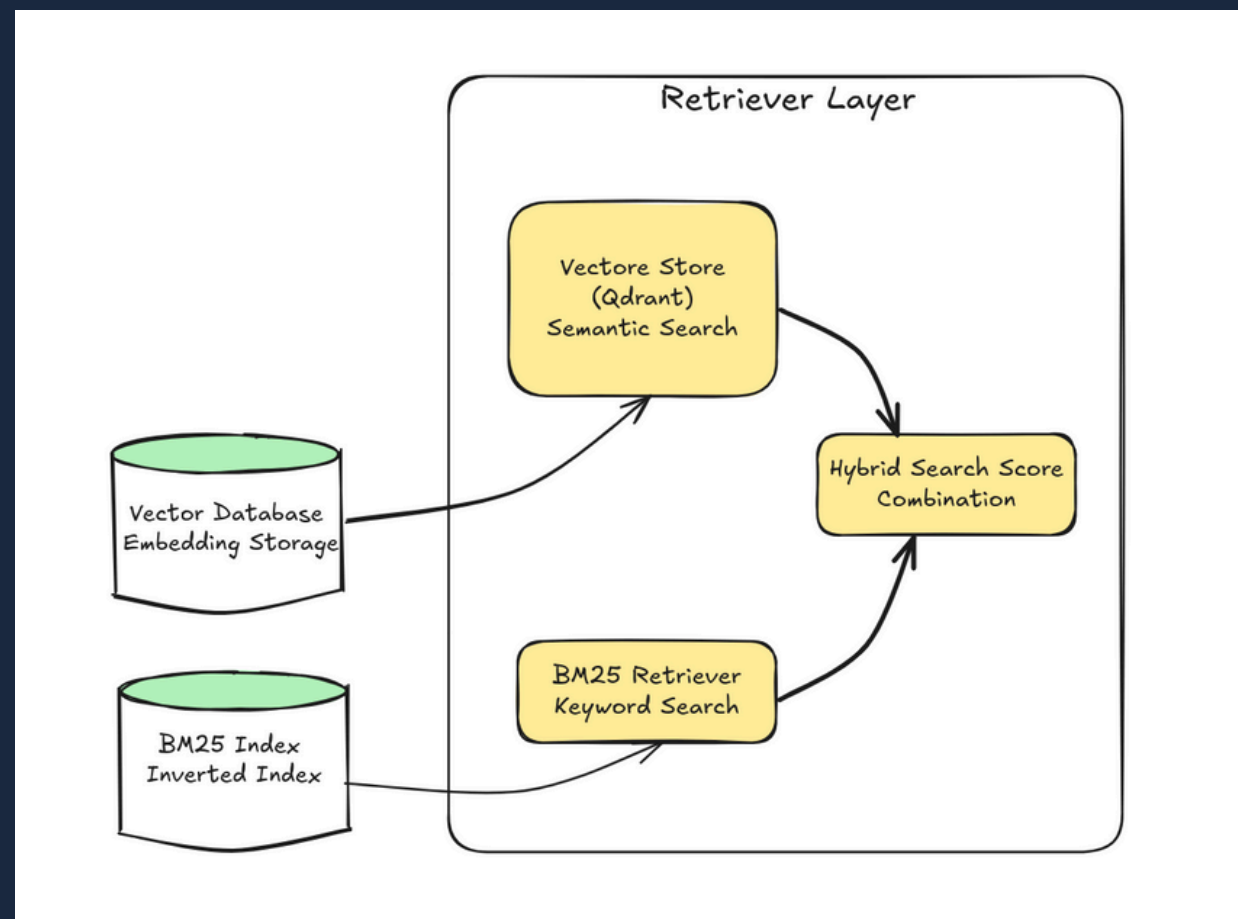


Figure 2.4.4. Hybrid Search

BM25: Regarding keyword search, methods such as TF-IDF and BM25 are commonly used to identify the most relevant documents based on term frequency in both the query and the documents.

After hyperparameter tuning (random search) chose $k1 = 1.2$ & $b = 0.65$

Limitations:

- Requires exact token match
- Ignores semantic meaning of words

The BM25 score for a document D and a query Q is calculated as follows:

$$\text{score}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgl}}\right)}$$

Where:

- $f(q_i, D)$ is the frequency of term q_i in document D
- $|D|$ is the length of document D
- avgl is the average document length in the collection
- $\text{IDF}(q_i)$ is the inverse document frequency of term q_i , calculated as:

$$\text{IDF}(q_i) = \ln \left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right)$$

With:

- N : total number of documents in the collection
- $n(q_i)$: number of documents containing the term q_i

2.4. MAIN FEATURES



HYBRID SEARCH

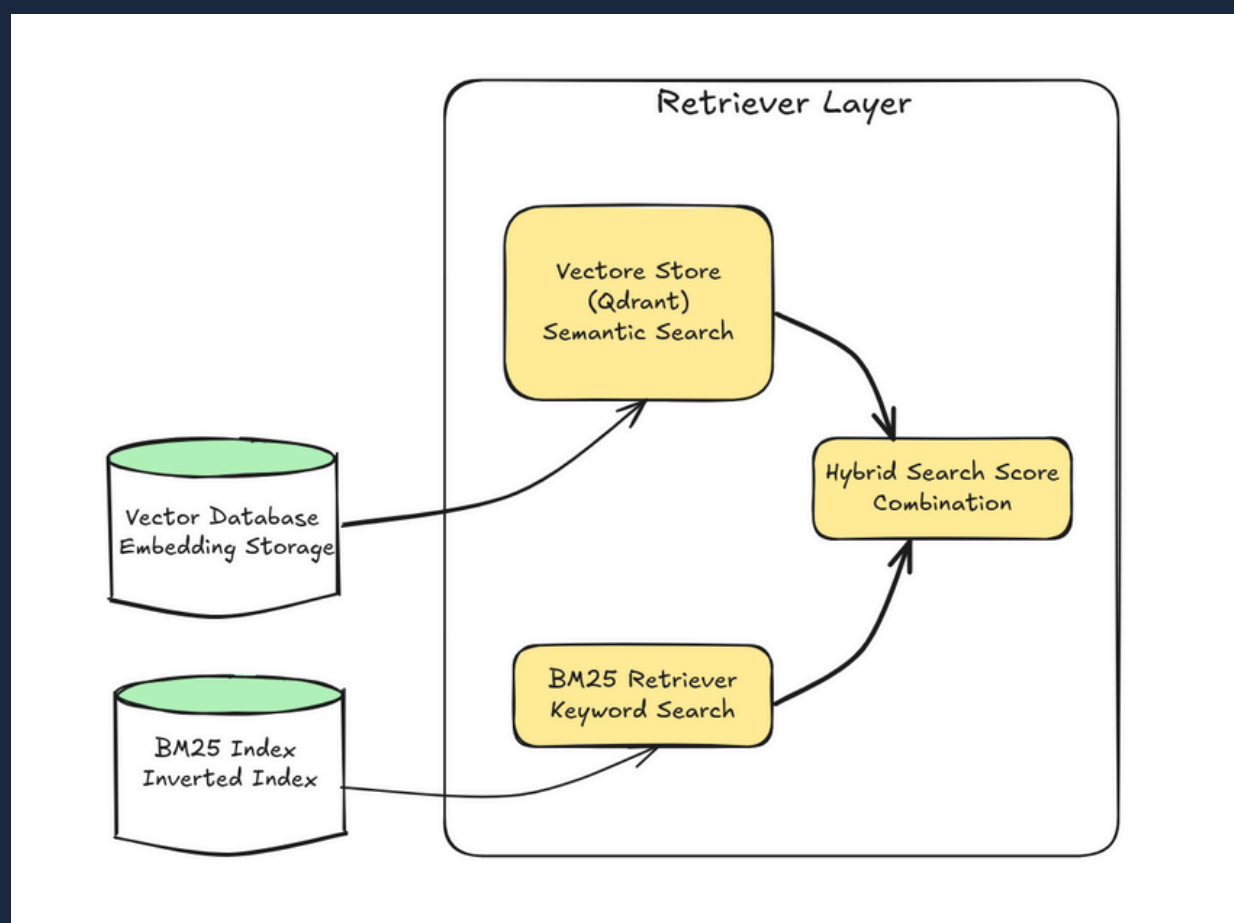


Figure 2.4.4. Hybrid Search

To address the aforementioned shortcomings of the BM25 search algorithm => **Vector Search**: Finds documents similar in meaning, even if different words are used.

In this implementation: compare 2 embedding models:
(sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2)
and
(bkai-foundation-models/vietnamese-bi-encoder)

2.4. MAIN FEATURES



HYBRID SEARCH

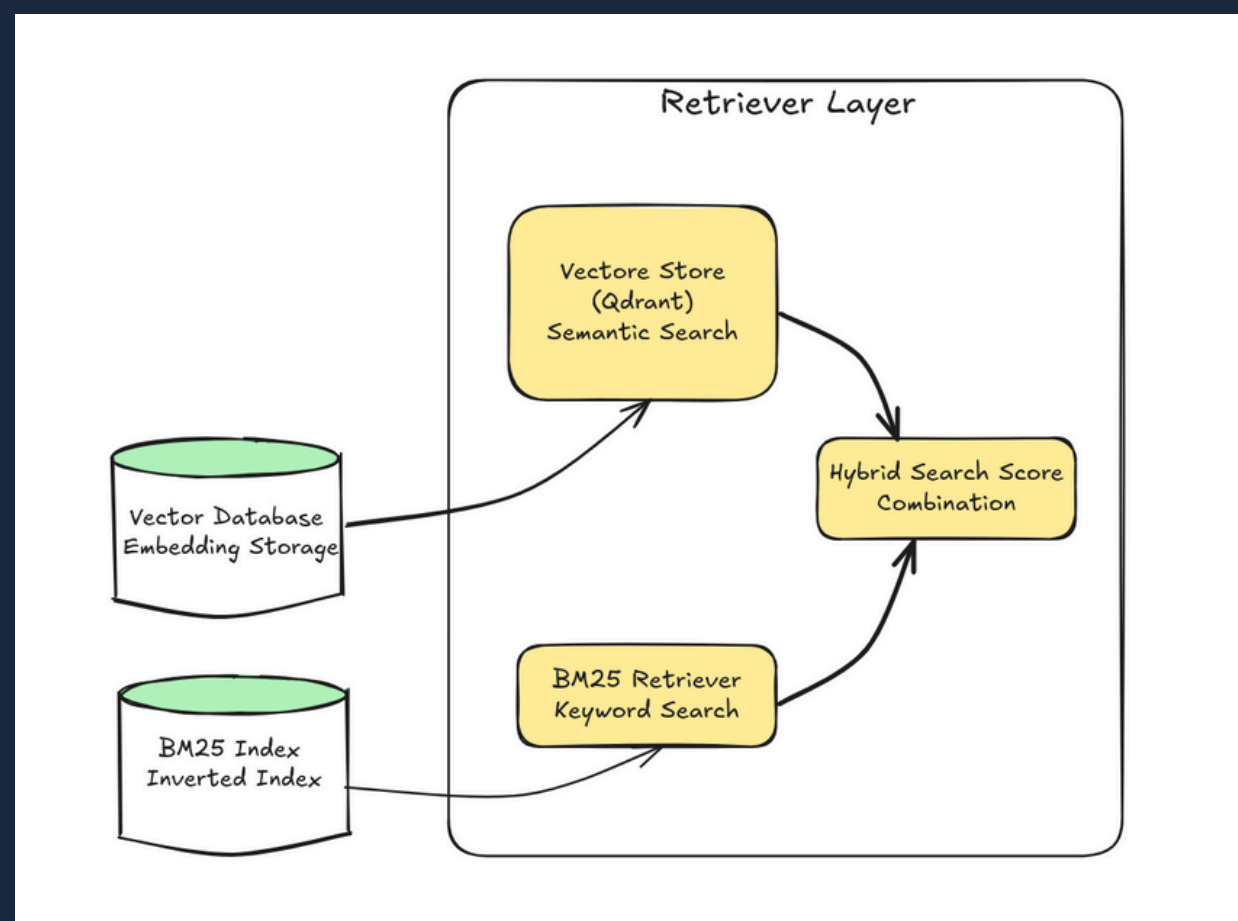


Figure 2.4.4 Hybrid Search

Note that the ordering of articles here is also based on score-based ordering of the chunks.

Hybrid Search (Combining Keyword & Semantic Retrieval): Though dense embeddings (semantic search) often outperform keyword-based methods in many tasks, we found that:

- For long or ambiguous legal documents, encoding models can lose effectiveness.
- Keyword search (e.g., BM25) serves as a valuable fallback.

Score Normalization & Fusion Strategy:

To leverage strengths of both methods:

- Retrieved scores from BM25 and Semantic Search are normalized to [0, 1]:
- Scores are combined using equal weights (0.5) for balance.

$$\text{normalized_score} = \frac{\text{score} - \min(\text{scores})}{\max(\text{scores}) - \min(\text{scores})}$$

2.3. MAIN FEATURES



CROSS-ENCODER RERANKING

After hybrid search (vector + BM25), we will get up to 25 possible results

A Cross-Encoder Reranker is a deep learning model that jointly encodes the query and candidate document to compute a single relevance score. Unlike a bi-encoder (which embeds query and document separately), a cross-encoder takes both as input together and outputs a more accurate score.

1. Input

- Query: The user's search question.
- Documents: A list of candidate documents (already retrieved by BM25, vector search, or hybrid).

2. Preparation

- For each document: Combine the title and content into a single string.
- Truncate the text if it's too long (to fit model limits). => Create a pair: [query, document_text].

3. Scoring

- Pass all [query, document_text] pairs to the cross-encoder model (cross-encoder/ms-marco-MiniLM-L-6-v2).
- The model predicts a relevance score for each pair.

4. Update & Sort

- For each document:

Original retrieval scores (from BM25 or Vector Search)

Reranked scores (from the Cross-Encoder)

- Sort all documents by reranker_score (descending).



2.3. MAIN FEATURES



SCORE-FUSION FINAL RANKING

Score Fusion is a technique used to combine multiple ranking scores into a single, unified score to improve the overall ranking quality.

In this project, Score Fusion is used to merge:

- Original retrieval scores (from BM25 or Vector Search)
- Reranked scores (from the Cross-Encoder)

Why Use Score Fusion? - Because:

- BM25 is good at keyword matching.
- Vector Search captures semantic similarity.
- Cross-Encoder offers deep relevance, but is expensive to compute.

=>

Each has strengths and weaknesses, so fusing them gives better, balanced result

$$\text{Final Score} = \alpha \cdot \text{Reranker Score} + (1 - \alpha) \cdot \text{Retriever Score}$$

Sort all documents by fused_score (descending) => Take top 20 documents



2.4. MAIN FEATURES



FALLBACK MECHANISMS

When the legal document database fails to provide a sufficient answer, my system seamlessly integrates Google Search as a fallback to enhance robustness and ensure response completeness.

1. Primary Fallback – No Documents Retrieved

If no legal documents are found after hybrid retrieval:

- Trigger Google Search
- If results are found:
 - Generate a fallback answer from web content
 - Return as the final response

Else:

- Return an error message indicating no relevant info

2. Secondary Fallback – Poor Answer Quality

If documents were retrieved but the generated answer is unsatisfactory or too generic:

- Trigger web search
- If successful:
 - Merge internal context + web info
 - Generate an improved response
- If no useful results:
 - Return original answer with a disclaimer or warning

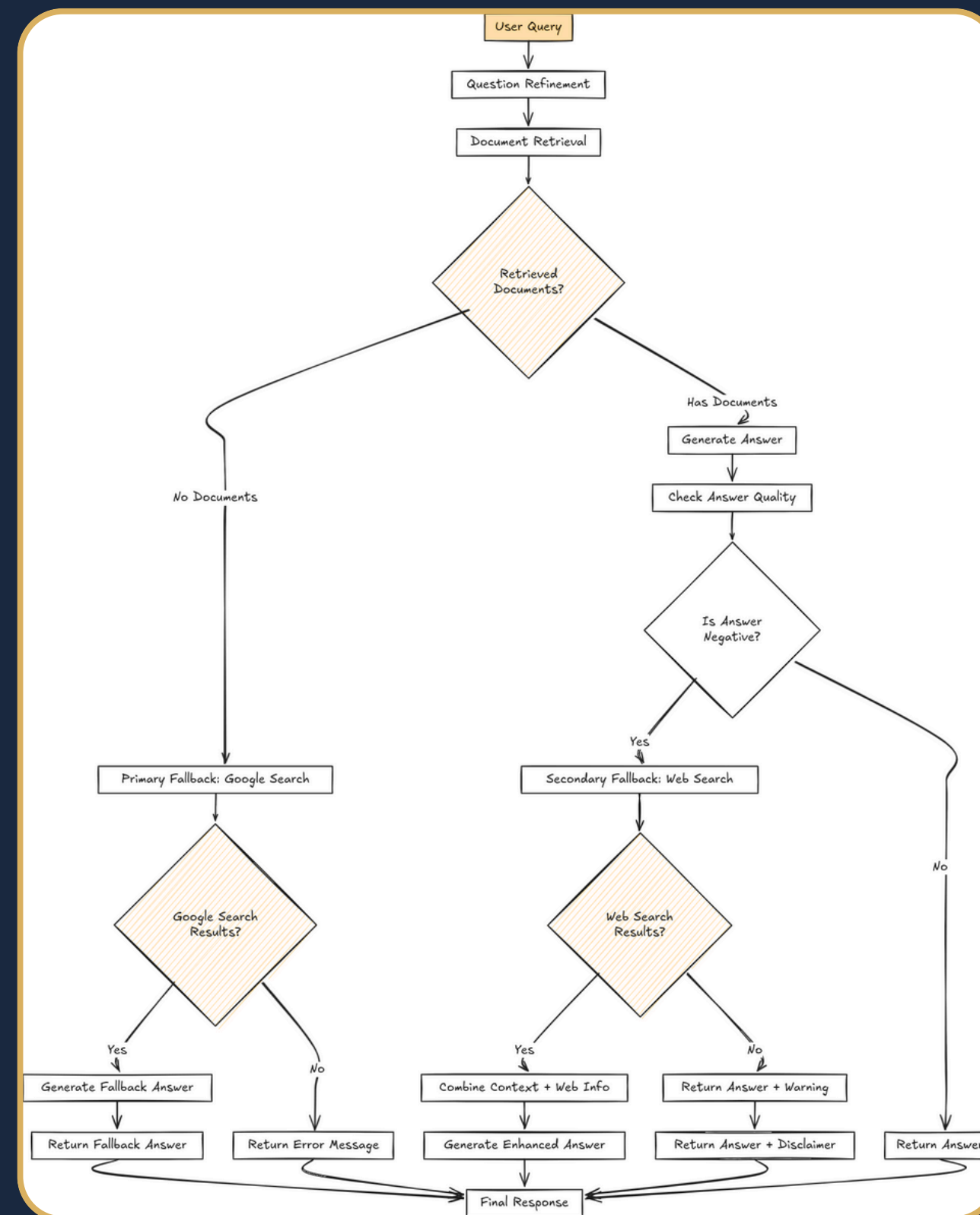


Figure 2.4.5. Fallback Mechanism

2.4. MAIN FEATURES



VIETNAMESE-OPTIMIZED

Most legal NLP systems are English-focused → This system is fully adapted for Vietnamese language and legal terminology.

Vietnamese NLP Optimization

Supports diacritic-preserving tokenization

Handles Vietnamese stopwords and common particles

Detects legal-specific patterns like Điều, Khoản, etc.

Legal AI Customization

Prompts tailored for Vietnamese legal questions

Legal-domain Bi-Encoder trained on local law corpus

Domain-aware reranking for more relevant results

Enhanced Legal Knowledge Search

Expands abbreviations (e.g., “BLDS” → Bộ luật dân sự)

Integrates Vietnamese law web search as fallback

Formats corpus with legal IDs and structured metadata

→ Fully adapted for Vietnamese language and legal system.

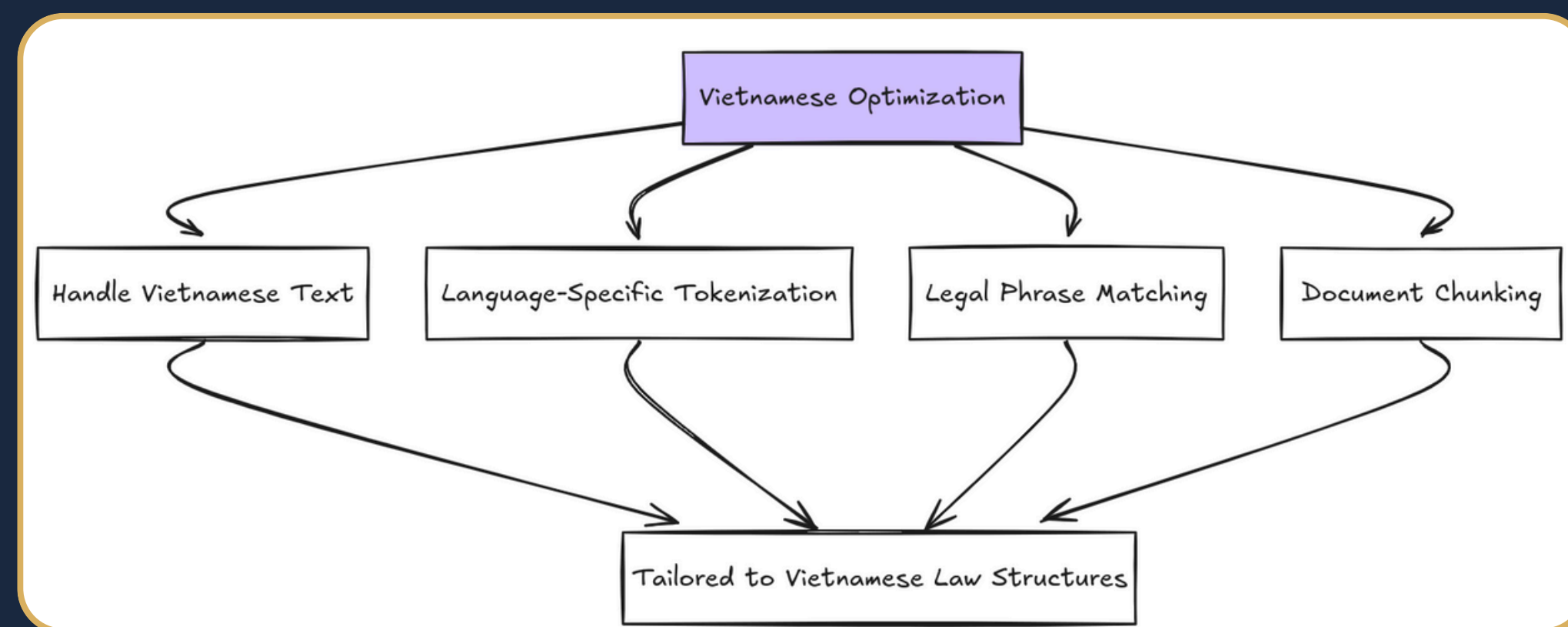


Figure 2.4.6. Vietnamese-optimized

03. USER STORIES & USE CASES

No	User Story	Content
1	Legal Professional Accessing Updated Legal Documents	1. As a legal professional, 2. I want to quickly access the most recent legal documents and updates, 3. So that I can stay informed about the latest changes in the law and provide accurate advice to my clients.
2	Law Student Researching Specific Legal Terms	1. As a law student, 2. I want to easily find definitions and contextual explanations of specific legal terms within Vietnamese legal documents, 3. So that I can deepen my understanding and support my academic research effectively.
3	Citizen Seeking to Understand Legal Procedures	1. As a concerned citizen, 2. I want a simple way to find and understand legal procedures and required forms related to personal or business matters, 3. So that I can confidently navigate administrative processes without needing professional legal help.

03. USER STORIES & USE CASES

No	User Story	Content
4	Researcher Analyzing Legal Trends	1. As a researcher specializing in Vietnamese law, 2. I want to analyze how the legal code has evolved over time, 3. So that I can produce insightful reports and contribute to academic or professional discussions.
5	User Providing Feedback for System Improvement	1. As a frequent user of the chatbot, 2. I want to provide feedback on the accuracy and relevance of the information I receive, 3. So that the system can continuously improve and better serve my future queries.
6	Individual Exploring Related Legal Queries	1. As someone interested in legal knowledge, 2. I want to receive suggestions for related legal questions when I search for a specific issue, 3. So that I can explore the broader legal context and gain a more comprehensive understanding.

03. USER STORIES & USE CASES

USE CASE 1: LAW SEARCH

Use case	Description
1. Search code of law	The primary action where a User searches for legal codes or documents.
2. Indexing Document	A process included in the search, likely involving organizing and preparing legal documents for searchability.
3. Search by keyword	An extension of the main search use case; allows searching based on specific keywords.
4. Search by category	Another extension of the main search use case; facilitates searching based on predefined legal categories.
5. Query from search database	Included in the process of Indexing Document; involves retrieving data from a search database as part of the indexing process.

USE CASE 2: QUESTION AND ANSWER

Use case	Description
1. Make a question	The core action where a User poses a question.
2. Model retrieval	A process included in making a question; possibly involves retrieving relevant information or models to answer the question.
3. Query from search database	Included in the process of Model retrieval; entails retrieving data from a database to assist in answering the question.

04. EVALUATION

4.1. EVALUATION SETUP

To evaluate the system's effectiveness in retrieving relevant legal articles based on natural language queries, multiple retrieval strategies were applied:

- BM25: Traditional keyword-based retrieval.
- Vector DB 1: Semantic search using MiniLM model.
- Vector DB 2: Semantic search using a fine-tuned Vietnamese Biencoder.
- Hybrid Methods: BM25 combined with Vector DBs.
- Hybrid + Reranking: Hybrid retrieval followed by Cross-Encoder reranking.

Metrics Used:

- MRR (Mean Reciprocal Rank): How early the first correct result is found (0 to 1 — higher is better).
- Coverage (Query with at least one correct hit): How often the system gives at least one correct result .
- Recall@k (R@1, R@10, R@20): How many correct document appear in top-k
- MAP@k (Mean Average Precision): How many correct answers are in the top k, and how early they appear.
- Time (s): Average retrieval time per query: How long the system takes to return an answer.



04. EVALUATION

Method	MRR	Coverage	Time (s)	R@1	R@10	R@20	MAP@1	MAP@10	MAP@20
BM25	0.5545	0.7894	0.183	0.430	0.768	0.783	0.430	0.562	0.565
Vector DB 1 (paraphrase-multilingual-MiniLM-L12-v2)	0.2465	0.4080	0.220	0.176	0.393	0.403	0.176	0.247	0.249
Vector DB 2 (vietnamese-bienocoder)	0.4691	0.6809	0.223	0.364	0.666	0.673	0.364	0.450	0.471
Hybrid 1 (BM25 + DB1)	0.5399	0.8282	0.425	0.430	0.761	0.821	0.430	0.541	0.550
Hybrid 2 (BM25 + DB2)	0.5801	0.8820	0.426	0.431	0.833	0.875	0.430	0.584	0.592
Hybrid 1 + Reranking	0.5864	0.8608	0.531	0.465	0.800	0.855	0.465	0.590	0.602
Hybrid 2 + Reranking	0.6082	0.8899	0.613	0.482	0.827	0.884	0.481	0.611	0.624

The Hybrid 2 + Reranking approach provided the best trade-off between accuracy and coverage, achieving:

- Highest MRR (0.6082): The first correct answer appears earliest
- Best Coverage (88.99%): In almost 9 out of 10 questions, the system gives at least one good answer.
- Best MAP@10 (0.611), MAP@20 (0.624): Among the top 10, 20 results, there are multiple good answers, and they appear early.
- Slightly increased latency (~0.61s/query), still acceptable for legal chatbot use.

05. CONCLUSION

This graduation project demonstrates the feasibility and effectiveness of applying modern AI techniques—particularly Retrieval-Augmented Generation (RAG)—to the Vietnamese legal domain. Despite limited resources and the challenges of working independently, I successfully developed a chatbot that combines keyword and semantic retrieval, supports user-friendly queries, and delivers accurate legal information. The system achieved strong performance, with a top configuration reaching an MRR of 0.6082 and Recall@10 of 0.827. More importantly, it highlights how technology can bridge the gap between legal knowledge and everyday users.

FUTURE WORKS

📌 1. Multilingual Support

Expand beyond Vietnamese to serve bilingual or international users.

📌 2. Integration with Official Databases

Connect to government legal databases for real-time updates and higher trustworthiness.

📌 3. Automatic Summarization

Enable concise summaries of lengthy or complex legal documents.

📌 4. Expert-in-the-Loop

Collaborate with legal professionals for feedback and quality assurance.

